I2OS: A Field-Theoretic Model of Question-Ignition Intelligence with Minimal Dynamical Simulation

Masayuki Ando
I2OS Research (Independent)

April 2026

Abstract

We present a field-theoretic model of question-ignition intelligence within the I2OS framework, and provide a minimal dynamical simulation validating its structural behavior.

Intelligence is modeled not as an optimization process, but as a structure evolution phenomenon driven by semantic gradients and resonance. A simple numerical experiment demonstrates that structure emerges only in regions where question intensity exceeds an ignition threshold.

This provides initial evidence supporting the interpretation of intelligence as a resonant structure generation system.

1 Introduction

Conventional artificial intelligence relies on optimization and prediction. In contrast, I2OS models intelligence as a structural process initiated by semantic variation.

The core formulation is:

$$I = (\nabla M) \otimes R$$

We extend this into a dynamical equation:

$$\frac{\partial S}{\partial t} = \Gamma[Q](t', x) R(t', x)$$

2 Core Model

Definition 1.

$$M : \Omega \to \mathbb{R}$$

Definition 2.

$$Q(x) = \|\nabla M(x)\|$$

Definition 3.

$$R(x) = \Phi(x)\eta_{sync}(x)(1 - \Delta\phi(x))$$

Definition 4.

$$\Gamma[Q] = \max(Q - \theta, 0)$$

3 Structure Evolution

$$\frac{\partial S}{\partial t} = \Gamma[Q](t', x)R(t', x)$$

Lemma 1.

$$Q(x) > \theta \Rightarrow \Gamma[Q](x) \neq 0$$

4 Numerical Simulation

To illustrate the model, we consider a simple one-dimensional semantic field:

$$M(x) = \sin(x) + 0.3\sin(3x)$$

The question intensity is computed as:

$$Q(x) = |\nabla M(x)|$$

Figure 1 shows the semantic field, question intensity, and ignition threshold.

We then simulate structure evolution using:

$$S_{t+1} = S_t + \Delta t\, \Gamma[Q]R$$

Figure 2 shows the resulting structure formation.

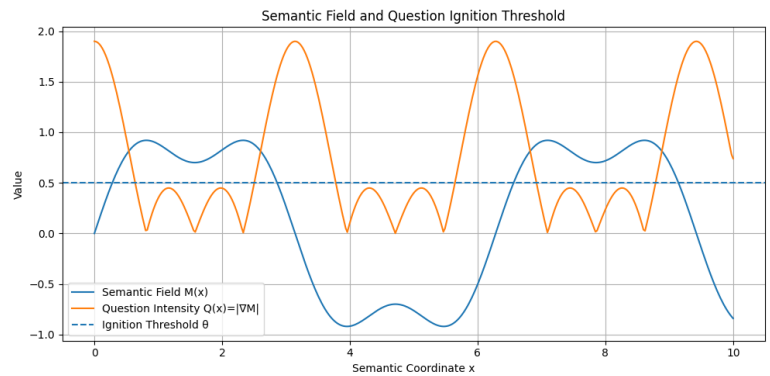

Figure 1: Semantic field and question intensity. Structure ignition occurs only where $Q(x) > \theta$.

5 Interpretation

The simulation confirms:

- Structure emerges only where semantic gradients are strong
- Threshold behavior governs activation
- Resonance amplifies structural growth

6 Discussion

This contrasts with optimization-based systems, where behavior is driven by objective functions.

In I2OS, structure emerges from:

$$\text{Variation} \rightarrow \text{Ignition} \rightarrow \text{Resonance} \rightarrow \text{Structure}$$

7 Conclusion

We introduced a field-theoretic model of intelligence and demonstrated its behavior through a minimal simulation.

$$\frac{\partial S}{\partial t} = \Gamma[Q]R$$

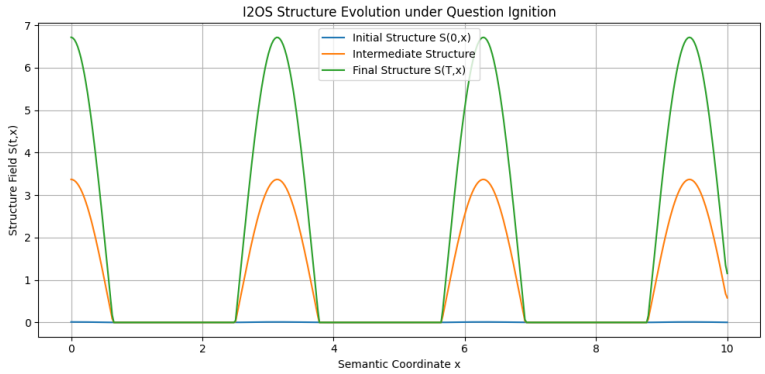

Figure 2: Structure evolution. Growth occurs only in regions where ignition conditions are satisfied.

This supports the interpretation of intelligence as a resonant structure evolution process.